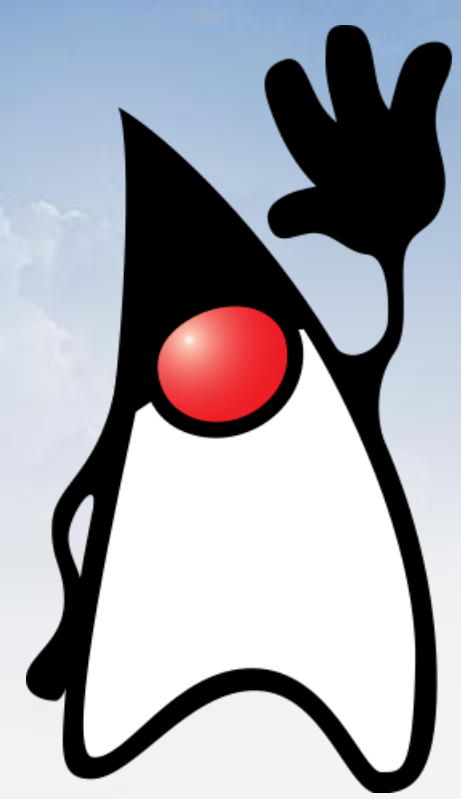




Java Intro

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Java

- developed by Sun Microsystems
 - Acquired by Oracle
- it is C++ --
 - Good stuff of C++ included
 - Bad stuff of C++ is missing
- Got very fast, close to C/C++
 - faster than Python, why?
- James Gosling, Ph.D.

https://en.wikipedia.org/wiki/James_Gosling



Write once run everywhere

- Java is platform independent
 - There is a virtual machine for every system
 - Code is written for the virtual machine not for the system
 - Java VM runs the code
- This gives flexibility, decreases the speed a little

JDK, JRE

- Java Runtime Environment
 - Necessary to run the java code
 - Contains libraries etc
 - has VM inside (java)
- JDK : Java Development kit
 - Necessary to compile the java code
 - Contains the java compiler (javac)
 - It also contains JRE

Compile vs Interpret

- Compile
 - Prepare the code for the machine
 - Optimize it
 - C/C++ are compiled.
- Interpretation
 - run the code line by line
 - NO optimization, or minimal
 - Python, Javascript, PHP are interpreted.

Compilation example

```
int i = 90;
int b = 10;
int result = 0;
i = i - b;
result++;
while (i >= b) {
    i = i - b;
    result++;
}
```

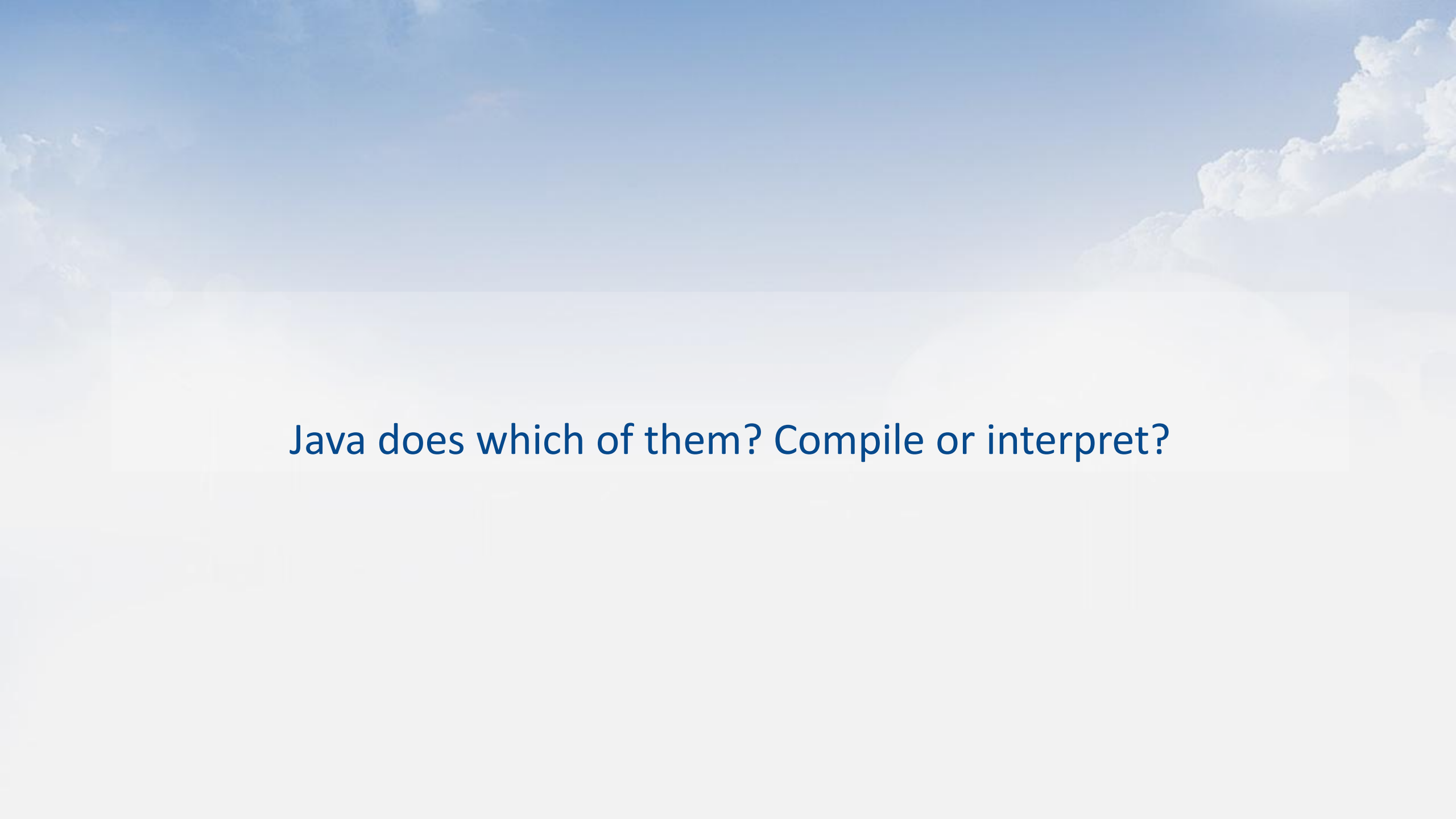
```
int i = 90;
int b = 10;
int result = 0;
while (i >= b) {
    i = i - b;
    result++;
}
```

i and b are
parameters,
coming from above

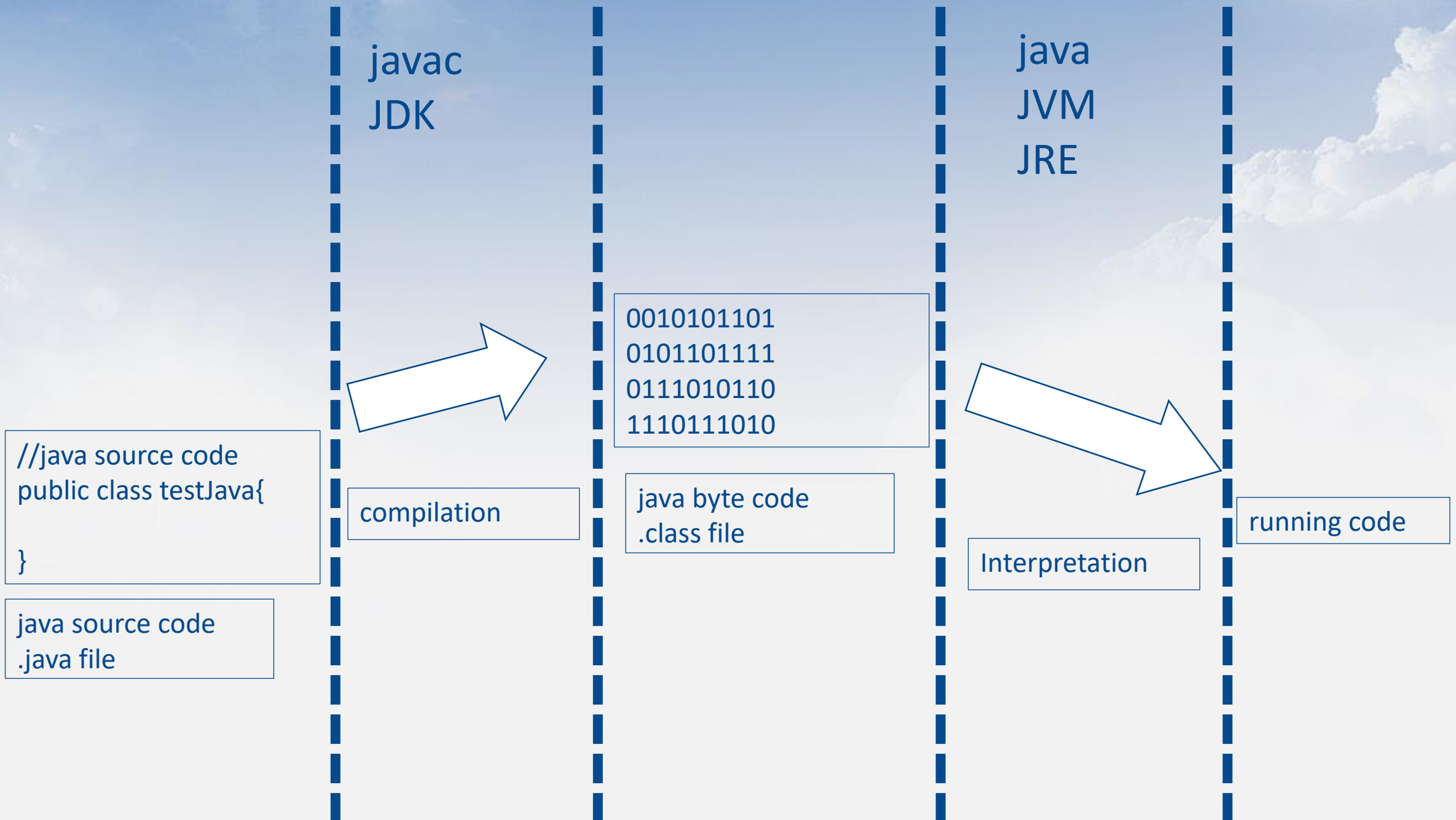
```
int i = 90;
int b = 10;
int result = i / b;
int remainder = i % b;
```

Always use
meaningful names
while naming
variables!



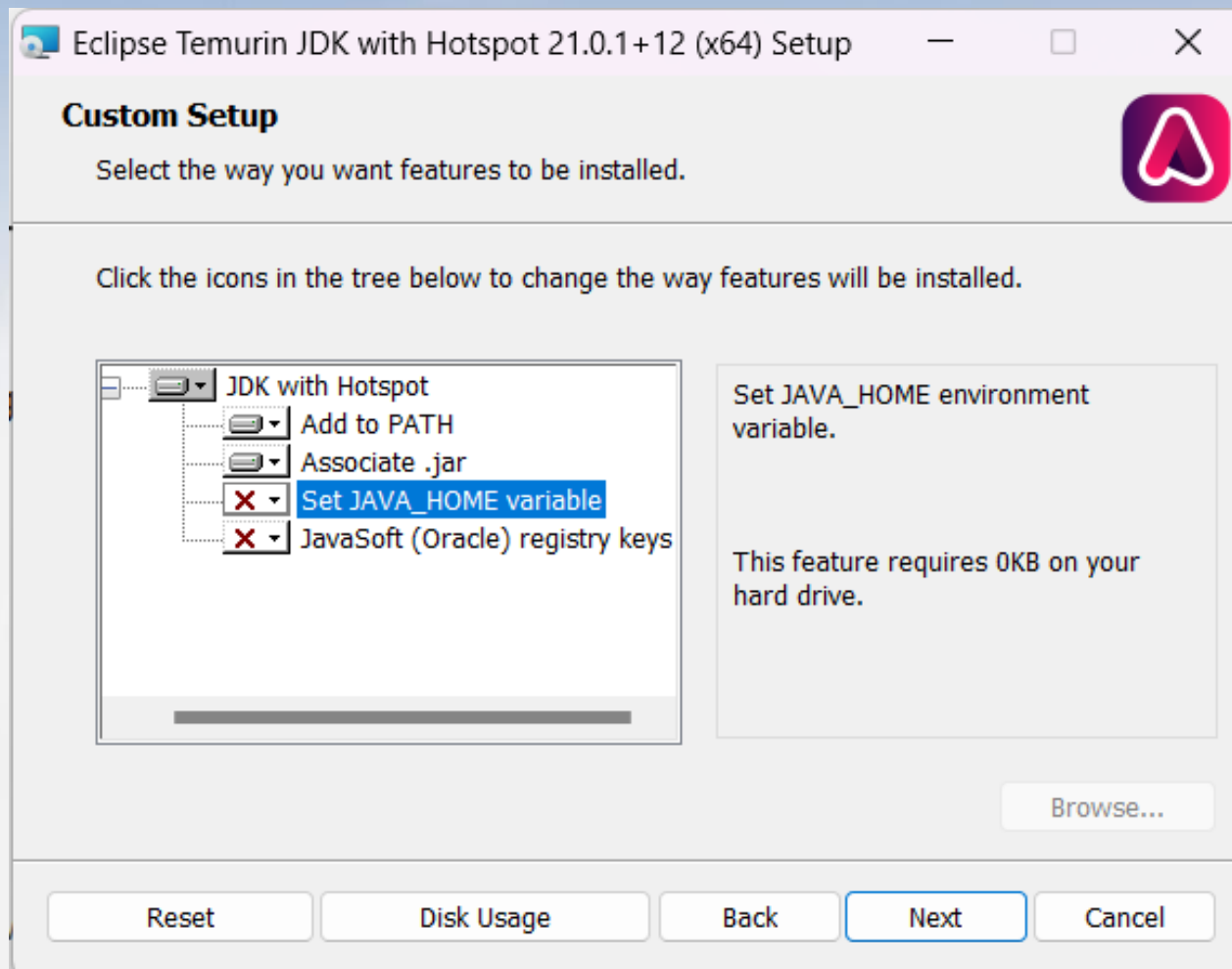
A background image of a bright blue sky with scattered white clouds. A semi-transparent white rectangular box is centered horizontally and vertically, containing the text.

Java does which of them? Compile or interpret?



Install Java Temurin

- <https://adoptium.net/temurin>



Dont touch
anything.
just next!

- Go to your command line.
- write javac
- You should be seeing this

```
Command Prompt
openjdk 21.0.1 2023-10-17 LTS
OpenJDK Runtime Environment Temurin-21.0.1+12 (build 21.0.1+12-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.1+12 (build 21.0.1+12-LTS, mixed mode, sharing)

C:\Users\Kara>javac
Usage: javac <options> <source files>
where possible options include:
  @<filename>                Read options and filenames from file
  -Akey[=value]              Options to pass to annotation processors
  --add-modules <module>(<module>)*
                             Root modules to resolve in addition to the initial modules,
                             or all modules on the module path if <module> is ALL-MODULE-PATH.
  --boot-class-path <path>, -bootclasspath <path>
                             Override location of bootstrap class files
  --class-path <path>, -classpath <path>, -cp <path>
                             Specify where to find user class files and annotation processors
  -d <directory>             Specify where to place generated class files
  -deprecation
                             Output source locations where deprecated APIs are used
  --enable-preview
                             Enable preview language features.
                             To be used in conjunction with either -source or --release.
  -encoding <encoding>       Specify character encoding used by source files
  -endorseddirs <dirs>        Override location of endorsed standards path
  -extdirs <dirs>             Override location of installed extensions
  -g                          Generate all debugging info
  -g:{lines,vars,source}      Generate only some debugging info
  -g:none                     Generate no debugging info
  -h <directory>
                             Specify where to place generated native header files
```



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Eclipse IDE for Java Developers

The essential tools for any Java developer, including a Java IDE, a Git client, XML Editor, Maven and Gradle integration



Eclipse IDE for Enterprise Java and Web Developers

Tools for developers working with Java and Web applications, including a Java IDE, tools for JavaScript, TypeScript, JavaServer...



Eclipse IDE for C/C++ Developers

An IDE for C/C++ developers.



Eclipse IDE for Embedded C/C++ Developers

An IDE for Embedded C/C++ developers. It includes managed cross build plug-ins (Arm and RISC-V) and debug plug-ins...

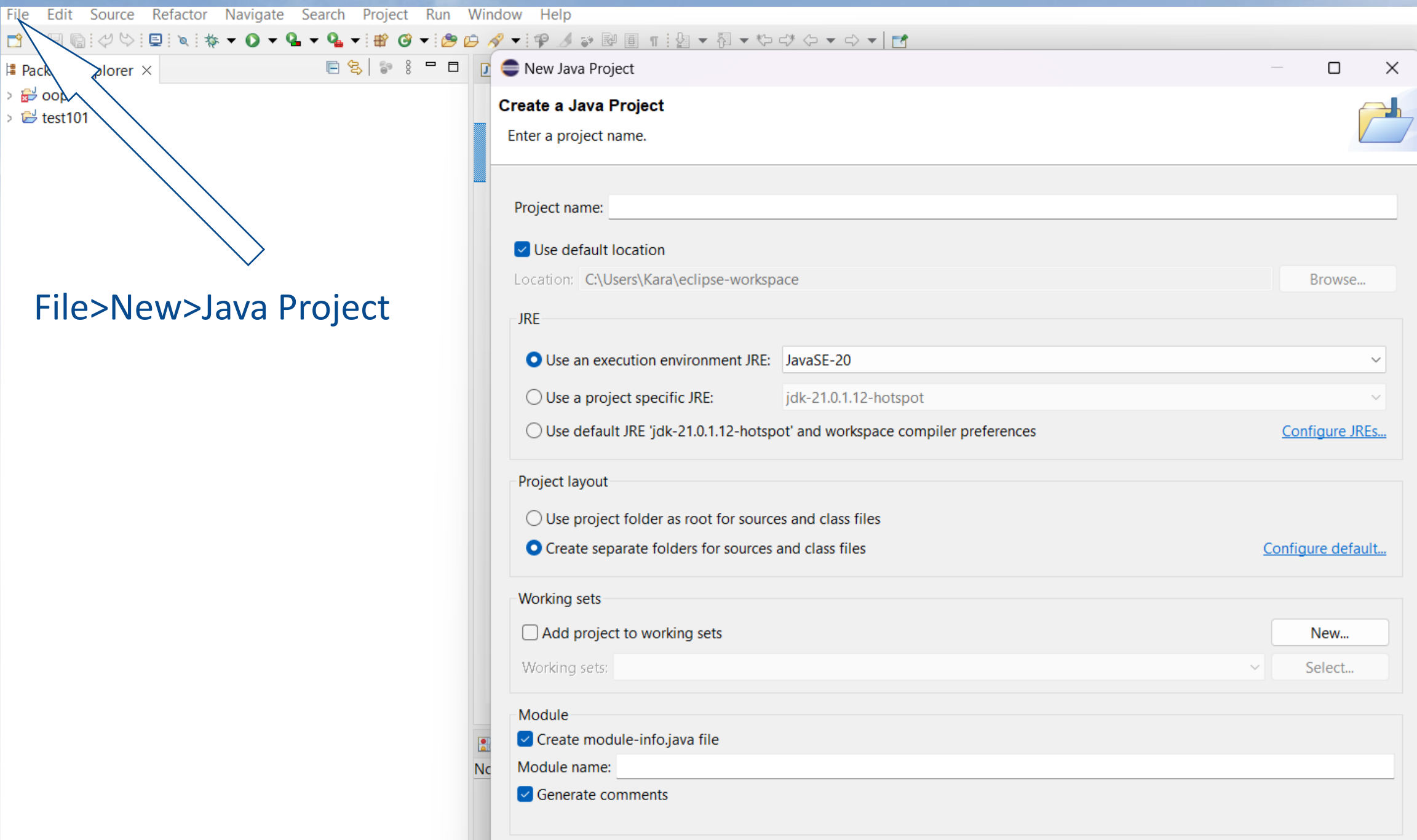


Eclipse IDE for PHP Developers

The essential tools for any PHP developer, including PHP language

If Java compiler is JDK what is Eclipse?

- Eclipse is just a shell to *javac* command, aka IDE
- Codeblocks is just a shell to *gcc*
- Rather than checking command line output and looking at our code in notepad
 - IDEs just give a nice User Interface
 - Package import options and compiler options are easier to edit



Assignments

- All assignments should be working with Temurin
 - Most probably you won't have a problem